Functional & Performance Testing Template

Model Performance Test

Date	21 February 2025
Team ID	EL2026TMID10104
Project Name	AI-Based Anomaly Detection Platform
Maximum Marks	

Test Scenarios & Results

Testing is an important phase in the software development lifecycle that ensures the system functions correctly and meets the specified requirements. The purpose of testing is to identify errors, validate system performance, and ensure reliability before deployment.

In this project, multiple test scenarios were executed to verify **functional correctness, data processing accuracy, API connectivity, and system performance**. Each test case includes a unique identifier, scenario description, testing steps, expected outcome, and the actual result.

Functional Testing

Functional testing ensures that the system behaves according to the defined requirements.

Test Case ID	Scenario (What to Test)	Test Steps (How to Test)	Expected Result	Actual Result	Pass/Fail
FT-01	Text Input Validation (e.g., dataset name, parameters)	Enter valid and invalid text in input fields	Valid inputs accepted, errors shown for invalid inputs	System validated inputs correctly	Pass
FT-02	Number Input Validation (e.g., threshold values, parameters)	Enter numbers within and outside the valid range	Accepts valid values and displays error for invalid inputs	System handled validation correctly	Pass
FT-03	Data Processing and Visualization	Provide dataset input and generate charts	Charts are generated correctly based on input data	Charts displayed successfully	Pass
FT-04	Database Connection Check	Verify connection between application and database	System successfully connects and retrieves data	Database connection successful	Pass

Performance Testing

Performance testing checks how the system behaves under different workloads and ensures that it operates efficiently.

Test Case ID	Scenario (What to Test)	Test Steps (How to Test)	Expected Result	Actual Result	Pass/Fail
PT-01	Response Time Test	Measure time required to process dataset and generate visualization	Processing time should be under 3 seconds	Response time within acceptable limits	Pass
PT-02	API Response Speed Test	Send multiple API requests simultaneously	System should respond without delay	System handled requests successfully	Pass
PT-03	File Upload Load Test	Upload multiple datasets and check processing performance	System processes files smoothly without crashing	System handled uploads efficiently	Pass

Testing Summary

The testing phase verified that all major functionalities of the system—including **input validation, data processing, visualization generation, database connectivity, and API responsiveness—operate as expected**. Performance tests confirmed that the system handles multiple operations efficiently without significant delays.

All test cases passed successfully, indicating that the system is stable and ready for deployment.